

Tuple

Another Data Type

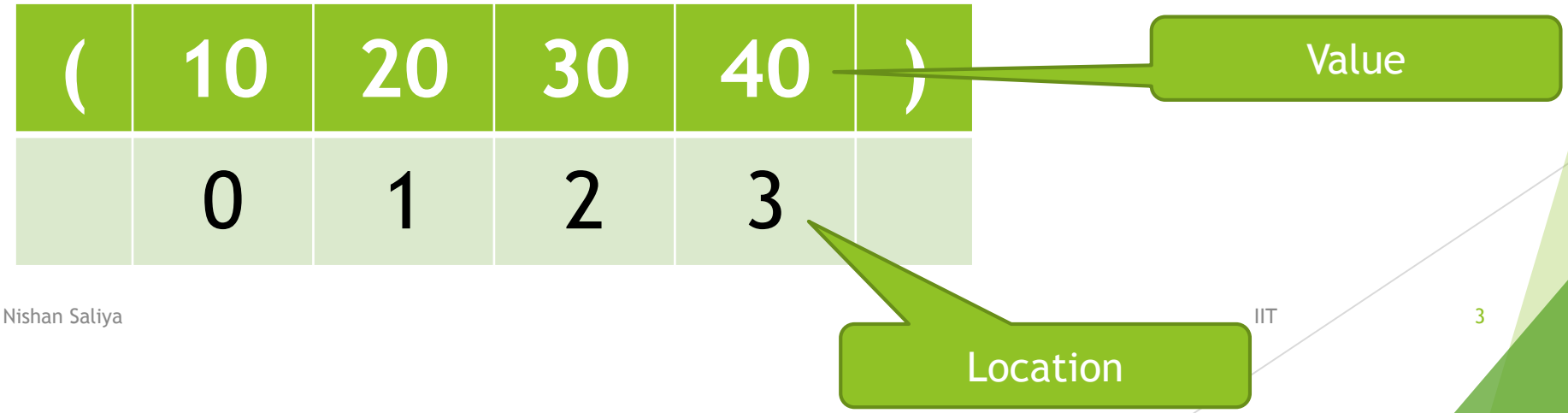
What is a Tuple?

- It’s a built-in data type
  - Just like a List
- Special structure to keep data
  - Immutable python object (No Updates)
- Use parentheses (not square brackets)
  - () Tuple
  - [] List
- Supports multiple data types

A Tuple

Take a look at the below sample codes for Tuples

- `MyTuple1 = ()`
- `MyTuple2 = (50,)`
- `MyTuple3 = (10, 20, 30, 40)`
- `MyTuple4 = (2017, "Python", "Cut-off", 50.00)`



Sample Tuple Program

```
# Create tuples
MyTuple1 = () # Empty tuple
MyTuple2 = (50,) # Single value tuple
MyTuple3 = (10, 20, 30, 40)
MyTuple4 = (2017, "Python", "Cut-off", 50.00)

# Display Tuples
print (MyTuple1)
print (MyTuple2)
print (MyTuple3)
print (MyTuple4)

# Display specific values (using slicer)
print (MyTuple2[0])
print (MyTuple3[2])
print (MyTuple4[1:3])
```

Update Tuple Elements

```
# Create tuples
MyTuple = (10, 20, 30, 40)

# Display Tuple
print (MyTuple)

# Update Tuple
MyTuple [2] = 60

# Display Tuple
print (MyTuple)
```

Create New Tuple

```
# Create tuples
Set1 = (1,2,3,4)
Set2 = (10,20,30,40)

print (Set1)
print (Set2)

# Join tuples
New = Set1 + Set2
print (New)

# Delete Tuples
del Set2

print (Set1)
print (Set2) # Possible ???
```

Simple Tuple Operations

```
# Create tuples
MyTuple = (10,20,30,40)

print (MyTuple)
print (len(MyTuple)) # Size
print (MyTuple * 2) # Two times
print (30 in MyTuple) # Is value there

# Dipslay with a loop
for Value in MyTuple:
    print (Value)
```

Convert List to a Tuple

```
# Create variables
MyList = [100, 200, 300, 400]
MyTuple = 0
print (MyList)
print (MyTuple)

# List -> Tuple
MyTuple = tuple(MyList)
print (MyTuple)
```

Convert Tuple to a List

```
# Create variables
MyTuple = ("Oh", "My", "God", "Python", "getting", "difficult")
MyList = 0

print (MyTuple)
print (MyList)

# Tuple -> List
MyList = list(MyTuple)
print (MyList)

# Updated to Son
MyList [2] = "Jesus"
print (MyList)
```